

Mini-Activity-2-Solutions.R

gcrone14

2026-10-05

Mini-Activity 2 Solutions

We start by setting up our environment.

```
# Load libraries
```

```
library(psych)
```

```
library(here)
```

```
## here() starts at /Users/gcrone14/Desktop/R Projects/PSYC3031-F26
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
```

```
## v dplyr      1.2.1      v readr      2.2.0
```

```
## v forcats    1.0.1      v stringr    1.6.0
```

```
## v ggplot2    4.0.3      v tibble     3.3.1
```

```
## v lubridate  1.9.5      v tidyr      1.3.2
```

```
## v purrr      1.2.2
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x ggplot2::%>%() masks psych::%>%()
```

```
## x ggplot2::alpha() masks psych::alpha()
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag() masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

1) Import the data into R and select the following variables and create a new dataframe with the following variables: ID, Extraversion, Neuroticism, Conscientiousness, Agreeableness, Openness, education, urban, age, race, married. You will use these eleven variables only in subsequent questions.

```
# Import data
```

```
personality <- read_csv(here::here("Data/TIPI_data.csv"))
```

```
## Rows: 1799 Columns: 17
```

```
## -- Column specification -----
```

```
## Delimiter: ","
```

```
## chr (9): education, urban, gender, engnat, religion, orientation, race, vote...
```

```
## dbl (8): ID, Extraversion, Neuroticism, Conscientiousness, Agreeableness, Op...
```

```
##
```

```
## i Use `spec()` to retrieve the full column specification for this data.
```

```
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

To check if data import worked, I'll just use `glimpse()`, but you can use whatever you wish.

```
# Check if data imported properly
glimpse(personality)
```

```
## Rows: 1,799
## Columns: 17
## $ ID <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ Extraversion <dbl> 3.5, 5.5, 4.0, 2.5, 2.0, 4.0, 3.5, 1.0, 4.5, 1.0, 1.~
## $ Neuroticism <dbl> 1.5, 7.0, 4.0, 2.5, 2.5, 3.5, 1.5, 1.0, 2.0, 6.5, 2.~
## $ Conscientiousness <dbl> 6.0, 4.5, 7.0, 5.0, 4.5, 4.5, 6.0, 3.0, 5.0, 1.0, 3.~
## $ Agreeableness <dbl> 6.0, 4.0, 3.0, 5.0, 2.5, 4.0, 4.5, 4.0, 6.5, 7.0, 1.~
## $ Openness <dbl> 6.5, 6.5, 7.0, 5.5, 4.0, 7.0, 5.0, 4.5, 6.0, 6.0, 4.~
## $ education <chr> "Univ", "<HS", "Univ", "HS", "HS", "HS", "<HS", "Uni~
## $ urban <chr> NA, "Suburban", "Rural", "Suburban", "Suburban", "Ru~
## $ gender <chr> "M", "F", "M", "M", "M", "M", "M", "M", "M", "F", "M~
## $ engnat <chr> "Non-native", "Native", "Native", "Native", "Native"~
## $ age <dbl> 28, 14, 25, 37, 17, 23, 17, 28, 32, 14, 16, 35, 18, ~
## $ religion <chr> "Atheist", "Agnostic", "Other", "Atheist", "Agnostic~
## $ orientation <chr> "Heterosexual", "Bisexual", "Heterosexual", "Bisexua~
## $ race <chr> "Other", "Indig-White", "Indig-White", "Indig-White"~
## $ voted <chr> "No", "No", "Yes", "No", "No", "No", "No", "Yes", "N~
## $ married <chr> "Never married", "Never married", "Never married", "~
## $ familysize <dbl> 1, 1, 3, 2, 2, 3, 2, 3, 3, 2, 2, 3, 3, 2, 5, 6, 3, 3~
```

```
# Select vars of interest
personality_select <- personality |>
  select(ID, Extraversion:Openness,
         education, urban, age,
         race, married)
```

2) Based on the type of variables in your new dataframe created in (1), identify those that may need to be converted to factors and do this conversion in R. Let's take a look at what variables we're working with.

```
glimpse(personality_select)
```

```
## Rows: 1,799
## Columns: 11
## $ ID <dbl> 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 1~
## $ Extraversion <dbl> 3.5, 5.5, 4.0, 2.5, 2.0, 4.0, 3.5, 1.0, 4.5, 1.0, 1.~
## $ Neuroticism <dbl> 1.5, 7.0, 4.0, 2.5, 2.5, 3.5, 1.5, 1.0, 2.0, 6.5, 2.~
## $ Conscientiousness <dbl> 6.0, 4.5, 7.0, 5.0, 4.5, 4.5, 6.0, 3.0, 5.0, 1.0, 3.~
## $ Agreeableness <dbl> 6.0, 4.0, 3.0, 5.0, 2.5, 4.0, 4.5, 4.0, 6.5, 7.0, 1.~
## $ Openness <dbl> 6.5, 6.5, 7.0, 5.5, 4.0, 7.0, 5.0, 4.5, 6.0, 6.0, 4.~
## $ education <chr> "Univ", "<HS", "Univ", "HS", "HS", "HS", "<HS", "Uni~
## $ urban <chr> NA, "Suburban", "Rural", "Suburban", "Suburban", "Ru~
## $ age <dbl> 28, 14, 25, 37, 17, 23, 17, 28, 32, 14, 16, 35, 18, ~
## $ race <chr> "Other", "Indig-White", "Indig-White", "Indig-White"~
## $ married <chr> "Never married", "Never married", "Never married", "~
```

Clearly, the categorical variables are education, urban, race, and married.

We should convert these to factors, as we do below.

I like to just remind myself in code what each level is.

```
table(personality_select$education) |> names()
```

```
## [1] "<HS" "Grad" "HS" "Univ"
```

```
table(personality_select$urban) |> names()
```

```
## [1] "Rural" "Suburban" "Urban"
```

```
table(personality_select$race) |> names()
```

```
## [1] "Arab" "Asian" "Black" "Indig-White" "Other"
```

```
table(personality_select$married) |> names()
```

```
## [1] "Currently married" "Never married" "Previously married"
```

To make things look a bit neater, I'll save character vectors to capture the unique levels of each categorical variable, then use those into the eventual conversion.

We should also really convert the ID variable to a character, so as not to get it confused with a numeric variable.

```
# Save unique levels of each variable
education_levels <- table(personality_select$education) |> names()
urban_levels <- table(personality_select$urban) |> names()
race_levels <- table(personality_select$race) |> names()
married_levels <- table(personality_select$married) |> names()

# Convert factor variables to factor type
personality_final <- personality_select |>
  mutate(ID = as.character(ID),
         education = factor(education, education_levels),
         urban = factor(urban, urban_levels),
         race = factor(race, race_levels),
         married = factor(married, married_levels))

# Check that modification worked
personality_final |>
  select(ID, education, urban, race, married) |>
  glimpse()
```

```
## Rows: 1,799
```

```
## Columns: 5
```

```
## $ ID      <chr> "1", "2", "3", "4", "5", "6", "7", "8", "9", "10", "11", "12~
```

```
## $ education <fct> Univ, <HS, Univ, HS, HS, HS, <HS, Univ, Univ, HS, HS, Grad, ~
```

```
## $ urban     <fct> NA, Suburban, Rural, Suburban, Suburban, Rural, Urban, Urban~
```

```
## $ race      <fct> Other, Indig-White, Indig-White, Indig-White, Indig-White, I~
```

```
## $ married   <fct> Never married, Never married, Never married, Currently marri~
```

3) Generate the appropriate descriptives (both numerical and graphical) for each individual variable. Let's start with the continuous variables.

```
personality_final |>
  select(Extraversion:Openness, age) |>
  psych::describe()
```

```
##           vars      n mean      sd median trimmed   mad   min   max
## Extraversion     1 1799  3.20   1.68    3.0    3.07   2.22   0.5   7.0
## Neuroticism      2 1799  3.97   1.76    4.0    3.97   2.22   0.5   7.0
## Conscientiousness 3 1799  4.39   1.56    4.5    4.44   1.48   1.0   7.5
## Agreeableness    4 1799  4.30   1.41    4.0    4.31   1.48   1.0   7.5
## Openness         5 1799  5.50   1.21    6.0    5.61   1.48   1.0   7.5
## age              6 1799 31.51 131.02   23.0   26.43  10.38  13.0 5555.0
##           range skew kurtosis   se
## Extraversion     6.5  0.54   -0.67 0.04
## Neuroticism      6.5 -0.01   -1.06 0.04
## Conscientiousness 6.5 -0.18   -0.79 0.04
## Agreeableness    6.5 -0.05   -0.51 0.03
## Openness         6.5 -0.80    0.24 0.03
## age             5542.0 41.65  1752.66 3.09
```

If we look at age, something is going wrong: there's someone in the data who is 5555 years old! Call Guinness World Records!

It's likely an error with how data were recorded, so I'd suggest putting an NA in its place instead.

```
personality_final <- personality_final |>
  mutate(age = ifelse(age > 200, NA, age))
```

Now let's take a look again

```
personality_final |>
  select(Extraversion:Openness, age) |>
  psych::describe()
```

```
##           vars      n mean      sd median trimmed   mad   min   max range
## Extraversion     1 1799  3.20   1.68    3.0    3.07   2.22   0.5   7.0   6.5
## Neuroticism      2 1799  3.97   1.76    4.0    3.97   2.22   0.5   7.0   6.5
## Conscientiousness 3 1799  4.39   1.56    4.5    4.44   1.48   1.0   7.5   6.5
## Agreeableness    4 1799  4.30   1.41    4.0    4.31   1.48   1.0   7.5   6.5
## Openness         5 1799  5.50   1.21    6.0    5.61   1.48   1.0   7.5   6.5
## age              6 1798 28.44  13.76   23.0   26.42  10.38  13.0  75.0  62.0
##           skew kurtosis   se
## Extraversion     0.54   -0.67 0.04
## Neuroticism     -0.01   -1.06 0.04
## Conscientiousness -0.18   -0.79 0.04
## Agreeableness    -0.05   -0.51 0.03
## Openness        -0.80    0.24 0.03
## age             1.14    0.42 0.32
```

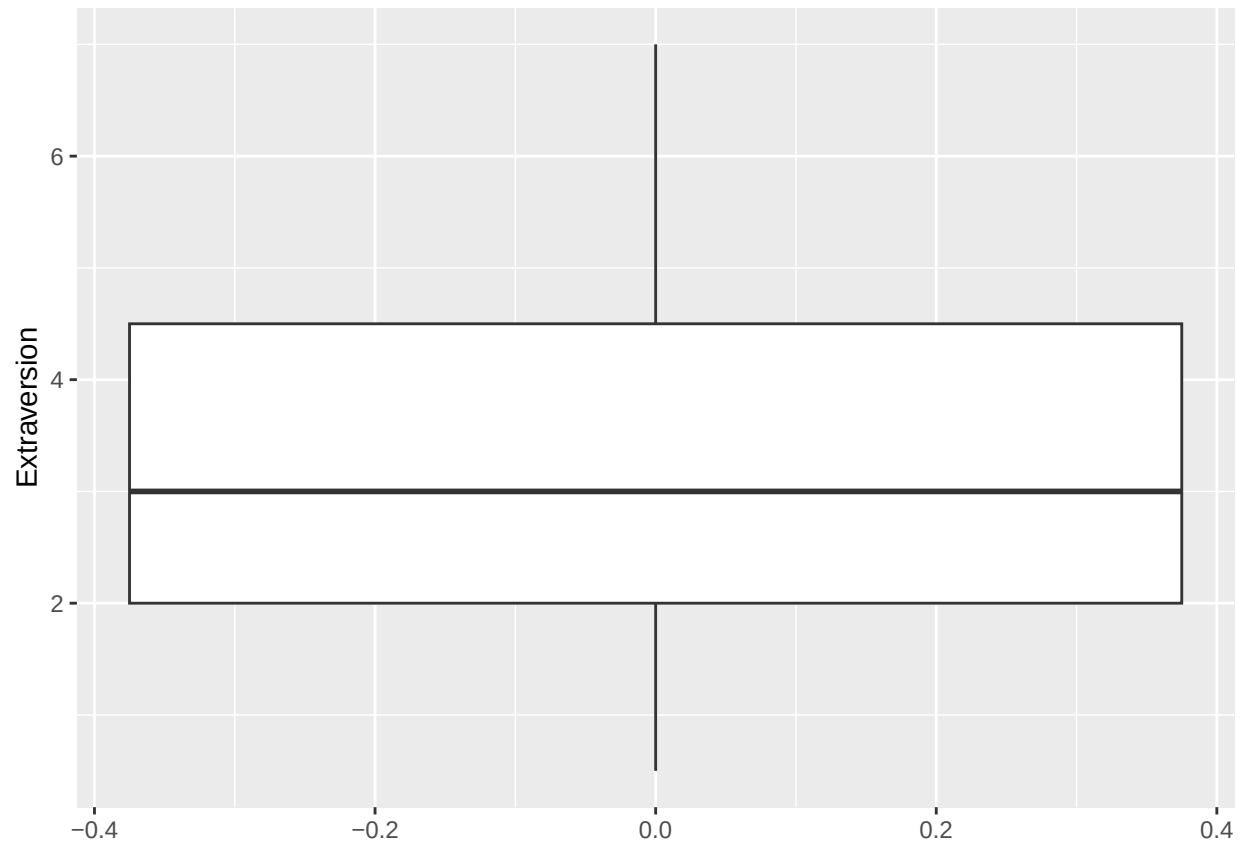
Age looks a lot better! Note that if you didn't repair age here, the boxplot for age would look extremely off and flattened. That's why it's so important to always explore your data!

If we want to plot each one, I'd simply make boxplots for each variable. The basic code is as follows:

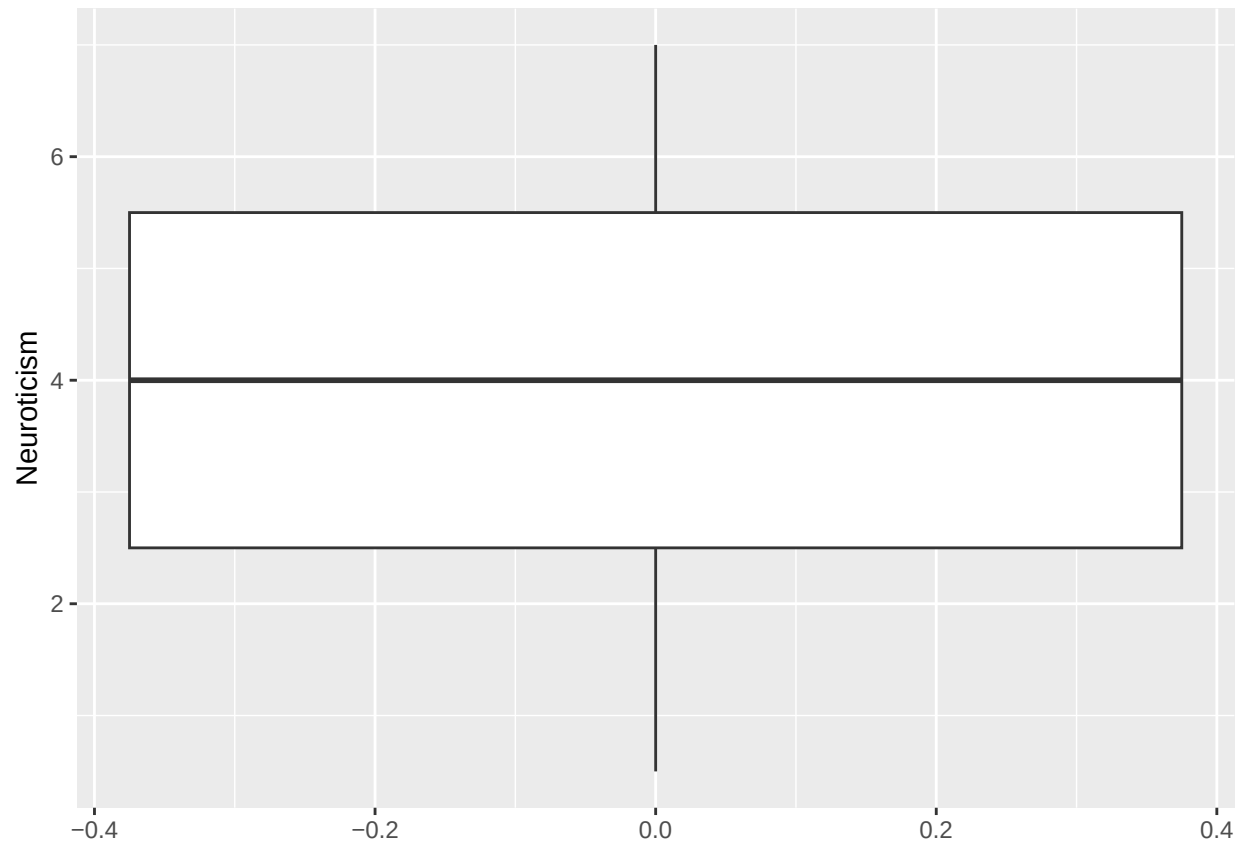
```
# personality_final |>
#   ggplot(aes(x = VARIABLE)) +
#   geom_boxplot()
```

Now we just sub in VARIABLE with each variable.

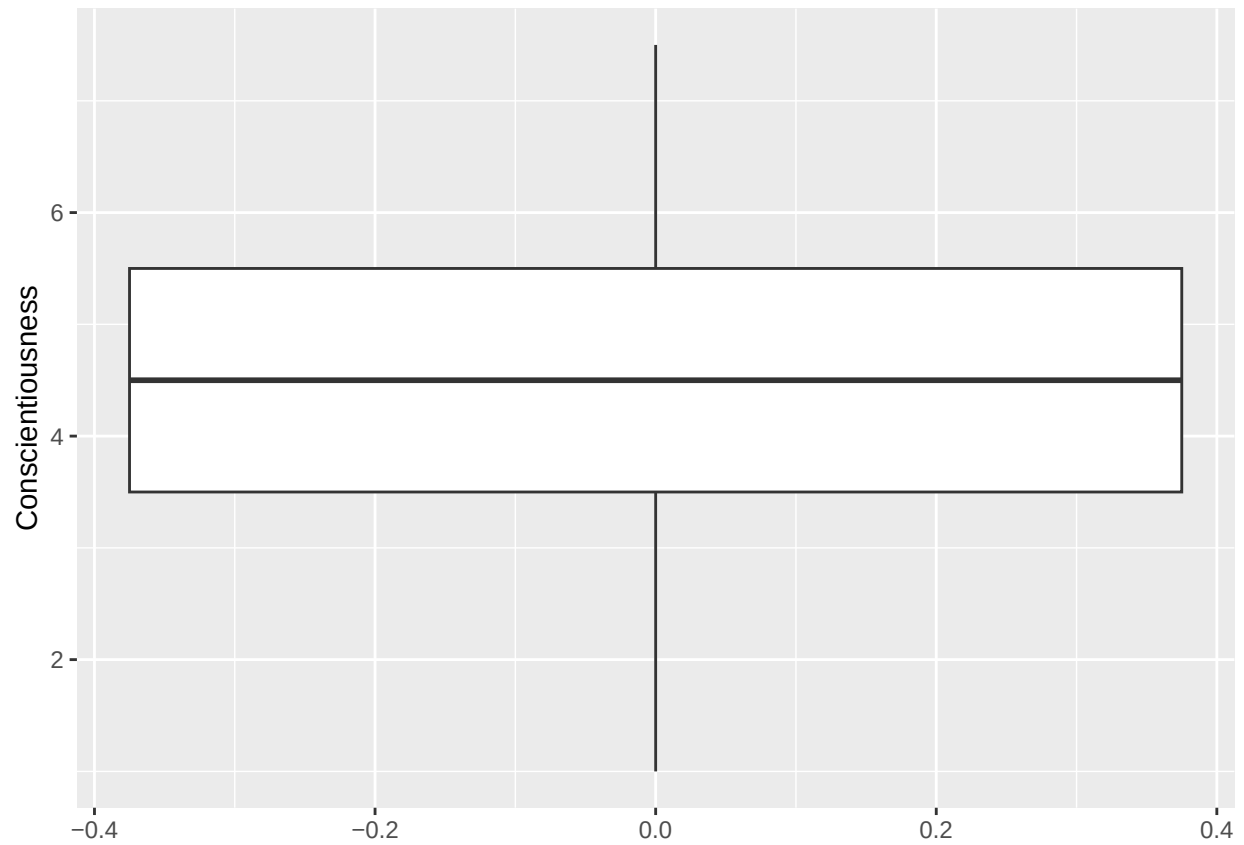
```
personality_final |>
  ggplot(aes(y = Extraversion)) +
  geom_boxplot()
```



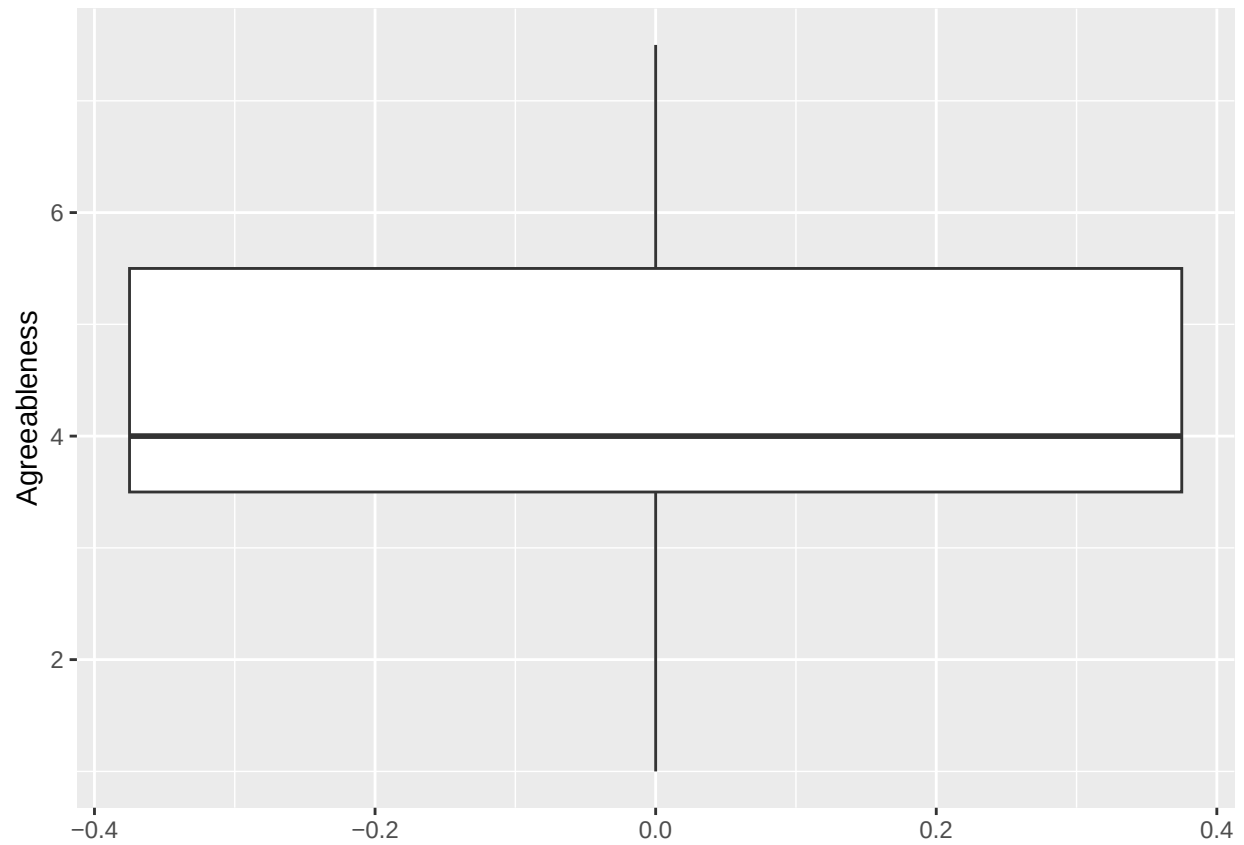
```
personality_final |>
  ggplot(aes(y = Neuroticism)) +
  geom_boxplot()
```



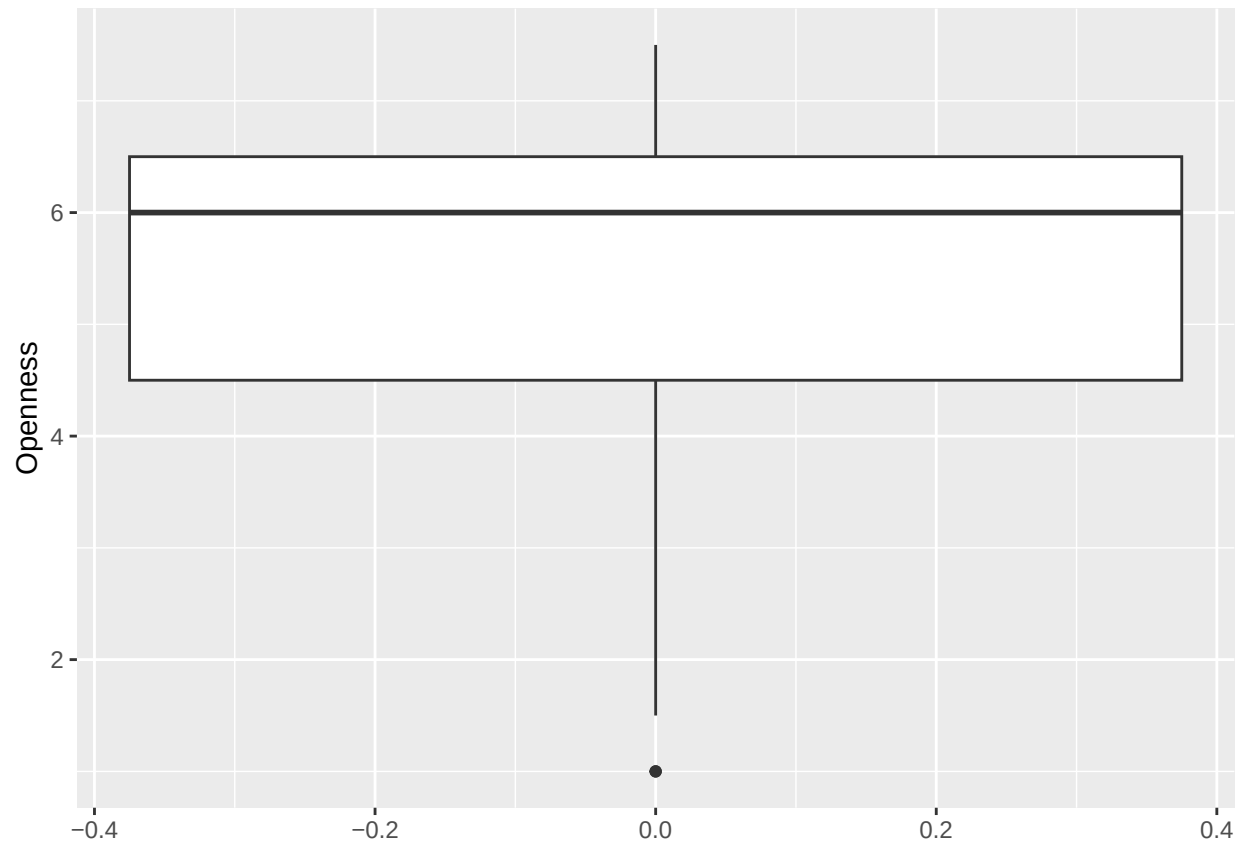
```
personality_final |>  
  ggplot(aes(y = Conscientiousness)) +  
  geom_boxplot()
```



```
personality_final |>  
  ggplot(aes(y = Agreeableness)) +  
  geom_boxplot()
```

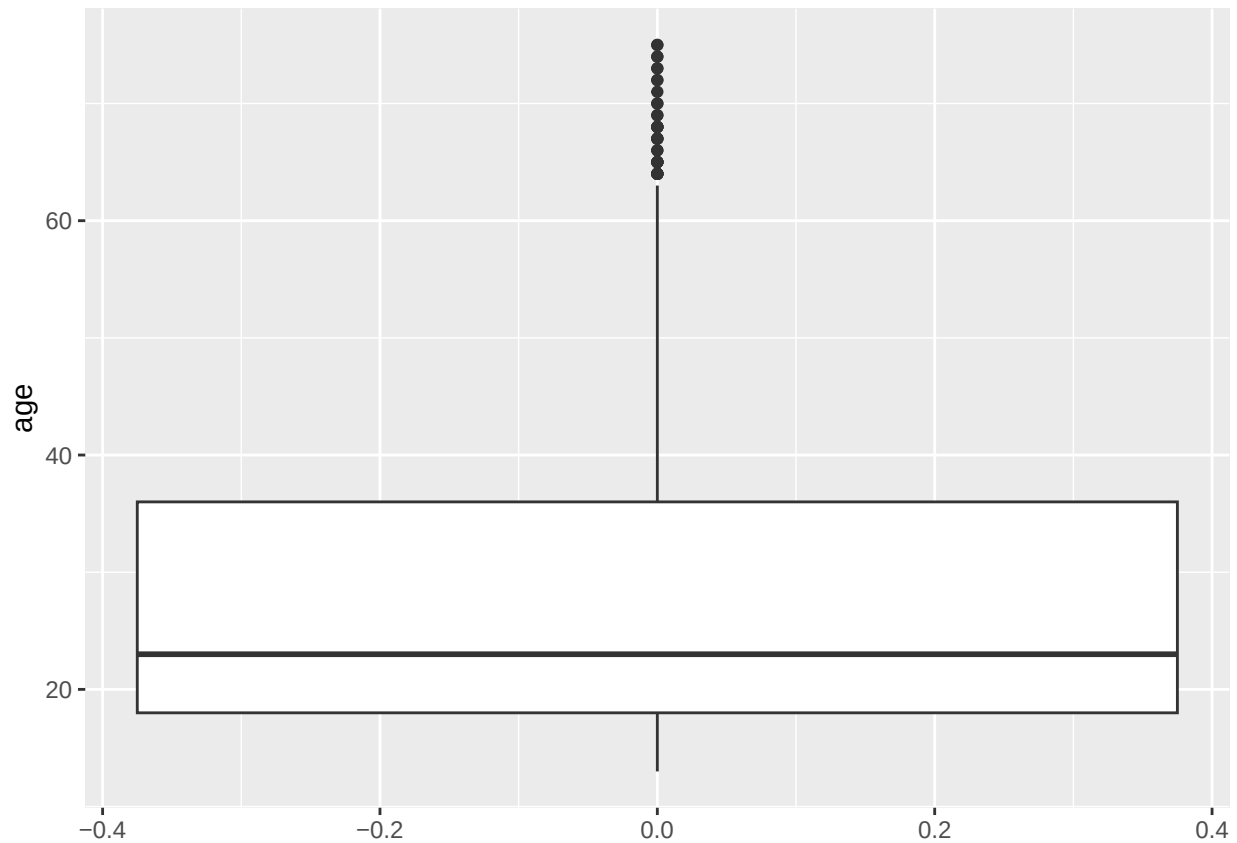


```
personality_final |>  
  ggplot(aes(y = Openness)) +  
  geom_boxplot()
```



```
personality_final |>  
  ggplot(aes(y = age)) +  
  geom_boxplot()
```

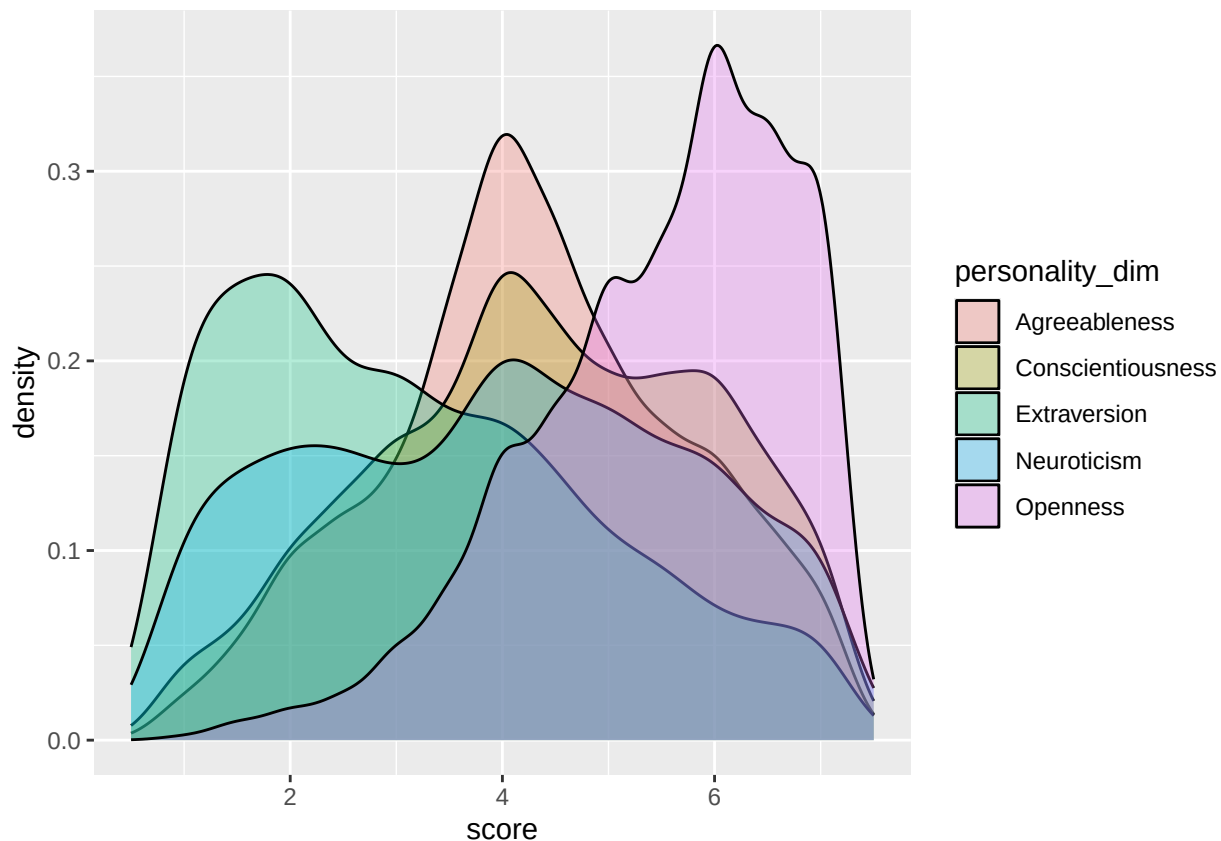
```
## Warning: Removed 1 row containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



All of the plots give us a nice overview of the data.

TOTALLY OPTIONAL: I'd find a density plot useful to see distributional differences on the personality characteristics.

```
personality_final |>
  # Advanced technique: convert data
  # to long-form to make later
  # plotting easier
  pivot_longer(cols = Extraversion:Openness,
    names_to = "personality_dim",
    values_to = "score") |>
  select(personality_dim, score) |>
  ggplot(aes(x = score, fill = personality_dim)) +
  geom_density(alpha = 0.3, adjust = 1)
```



Each personality characteristic has a unique density curve, each with a different central tendency, variability, and skew. (These are reflected in the descriptive stats from earlier.)

For the categorical variables, we simply generate frequency counts for each one. Quickest way to do this is using the `summary()` function.

```
personality_final |>
  select(education, urban, race, married) |>
  summary()
```

```
##  education      urban      race      married
##  <HS :278 Rural  :317 Arab      : 14 Currently married : 314
##  Grad:245 Suburban:815 Asian    : 119 Never married    :1383
##  HS :757 Urban   :645 Black    : 28 Previously married: 93
##  Univ:488 NAs    : 22 Indig-White:1440 NAs                : 9
##  NAs : 31
##                      Other    : 179
##                      NAs      : 19
```

We immediately get a sense of the frequency of each group in our data. The sample consists of mostly high school graduates, those living in suburbia, white people, and those who have never been married.

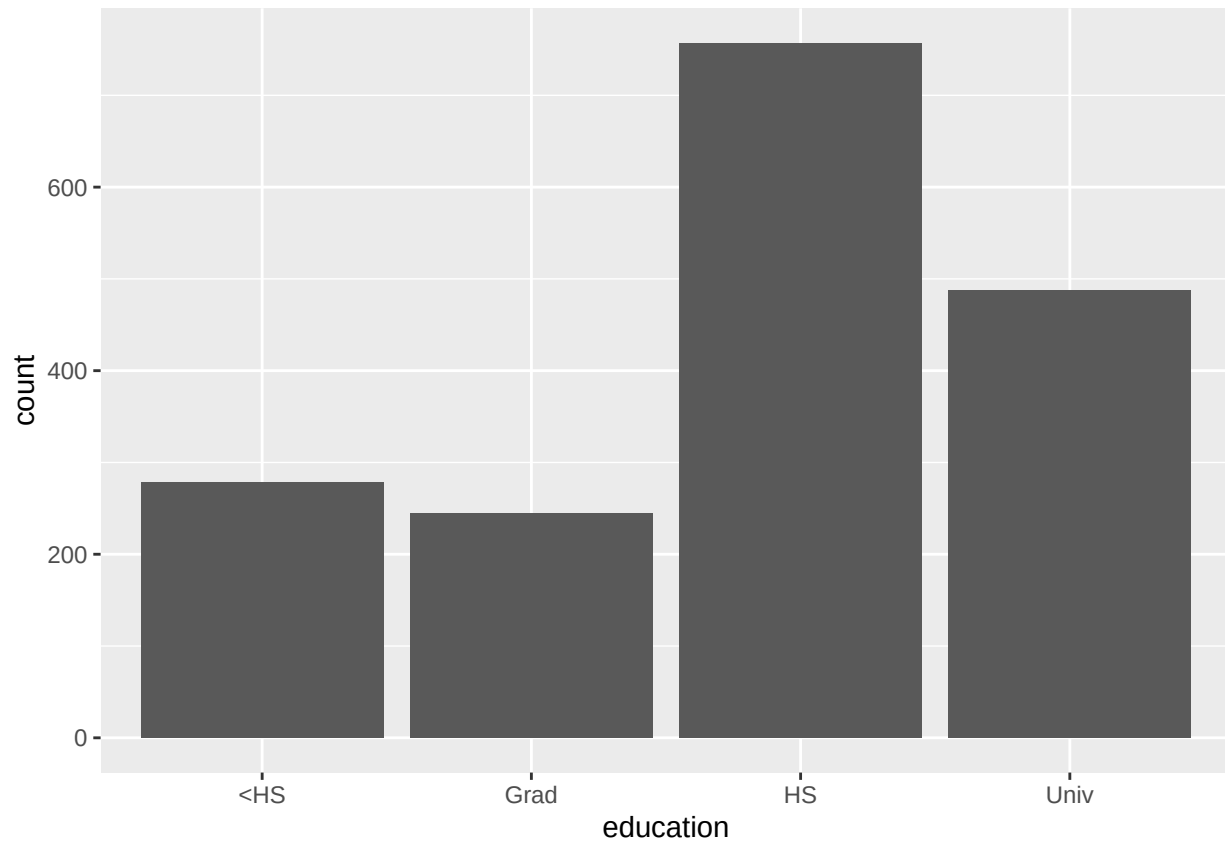
(This information is useful when considering how generalizable our results will be if using inferential tests.)

It's usually not worth it to make plots of this information, but if you had to, you can make simple bar plots.

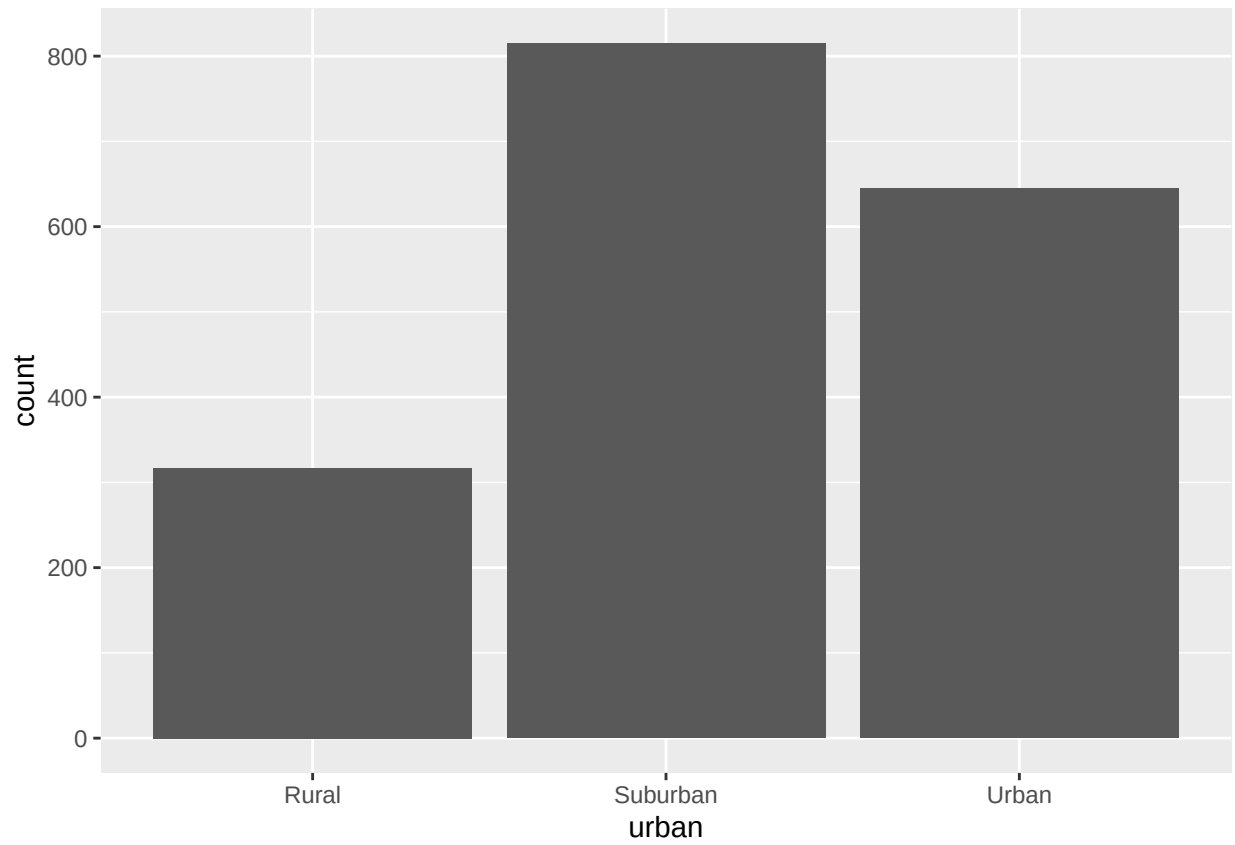
The generic formatting is as follows:

```
# personality_final |>
#   filter(!is.na(VARIABLE)) |>
#   ggplot(aes(x = VARIABLE)) +
#   geom_bar()
```

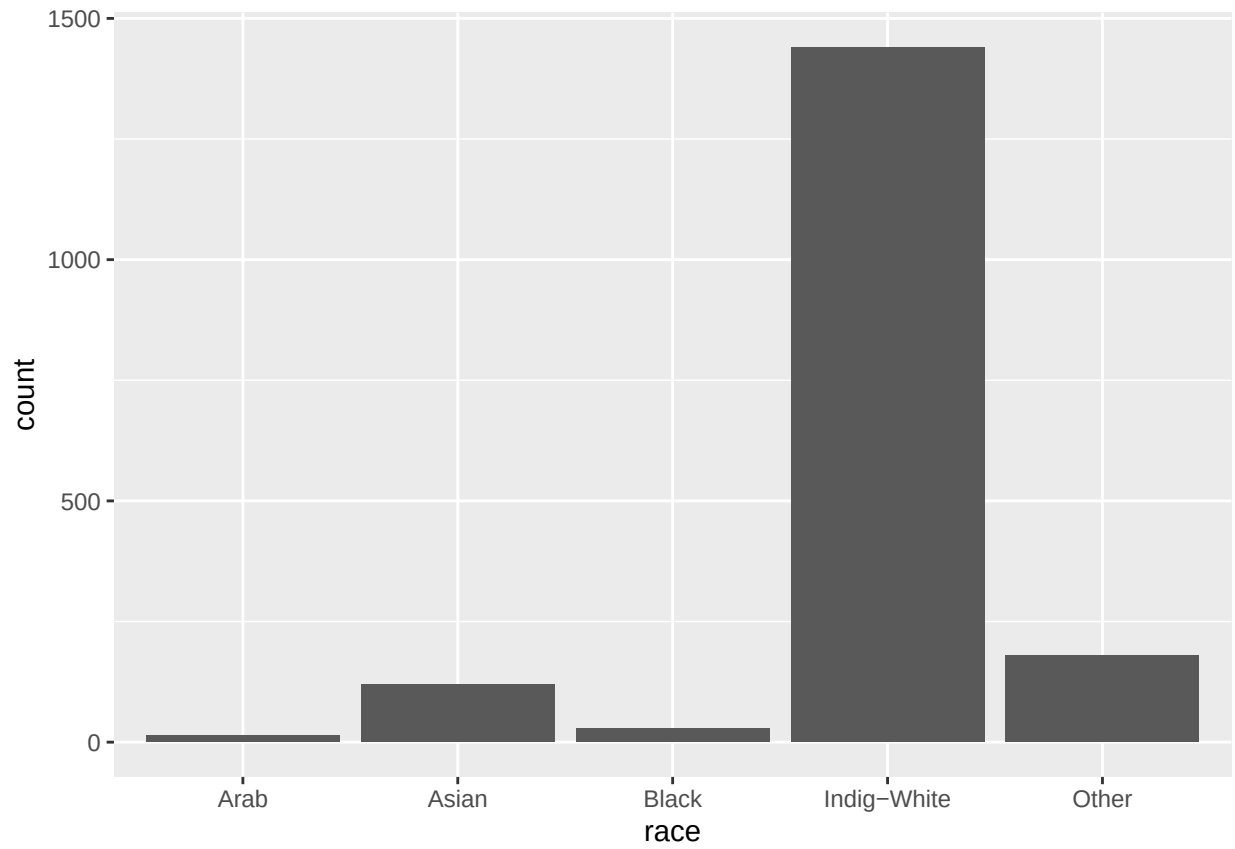
```
personality_final |>
  filter(!is.na(education)) |>
  ggplot(aes(x = education)) +
  geom_bar()
```



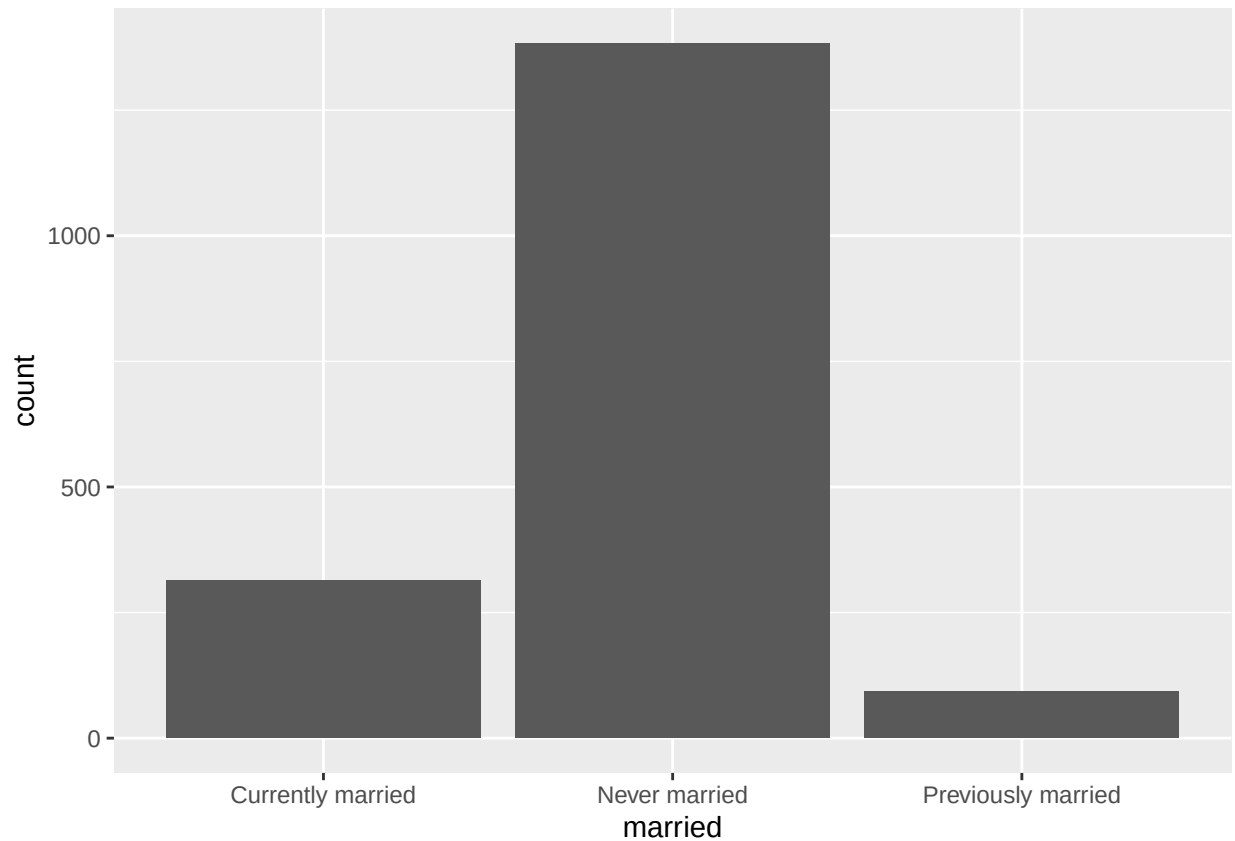
```
personality_final |>
  filter(!is.na(urban)) |>
  ggplot(aes(x = urban)) +
  geom_bar()
```



```
personality_final |>  
  filter(!is.na(race)) |>  
  ggplot(aes(x = race)) +  
  geom_bar()
```



```
personality_final |>  
  filter(!is.na(married)) |>  
  ggplot(aes(x = married)) +  
  geom_bar()
```



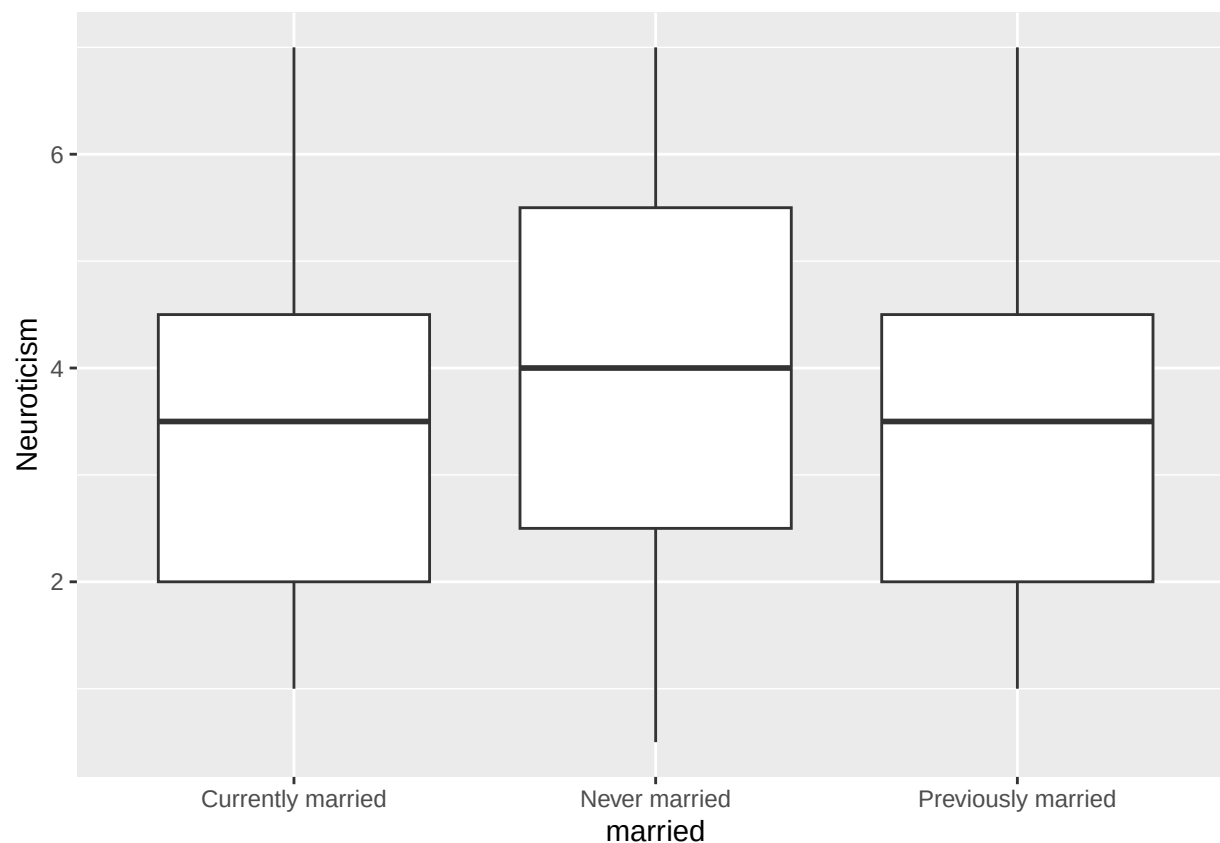
4) Choose two variables only and generate the appropriate graph to display their relationship.

This question will vary based on what you decide to check for.

I'm curious to see if neuroticism differs as a function of one's marital status. Let's take a look...

The plot we'd want to make is a boxplot, using marital status to separate each box.

```
personality_final |>
  # Filter NAs for clearer plot
  filter(!is.na(married)) |>
  ggplot(aes(x = married, y = Neuroticism)) +
  geom_boxplot()
```



There appear to be only slight differences between groups.

It'd help to see the numeric descriptives accompanying this graph.

```
psych::describeBy(personality_final$Neuroticism,
  personality_final$married)
```

```
##
## Descriptive statistics by group
## group: Currently married
##   vars   n mean   sd median trimmed  mad min max range skew kurtosis   se
## X1     1 314 3.47 1.65   3.5   3.41 2.22   1   7    6 0.25   -0.96 0.09
## -----
## group: Never married
##   vars   n mean   sd median trimmed  mad min max range skew kurtosis   se
## X1     1 1383 4.12 1.77    4   4.14 2.22 0.5   7   6.5 -0.11   -1.05 0.05
## -----
## group: Previously married
##   vars   n mean   sd median trimmed  mad min max range skew kurtosis   se
## X1     1  93 3.44 1.63   3.5   3.36 2.22   1   7    6 0.35   -0.66 0.17
```

More simple display

```
personality_final |>
  group_by(married) |>
  summarise(mean_neuro = mean(Neuroticism),
```

```
sd_neuro = sd(Neuroticism)) |>
# Drop out the NAs for clarity
drop_na() |>
# Arrange values from highest to lowest
arrange(desc(mean_neuro))
```

```
## # A tibble: 3 x 3
##   married      mean_neuro sd_neuro
##   <fct>          <dbl>    <dbl>
## 1 Never married      4.12      1.77
## 2 Currently married   3.47      1.65
## 3 Previously married  3.44      1.63
```

The mean neuroticism ($M = 4.12$, $SD = 1.77$) is highest within those who are never married. The mean for those who are currently married ($M = 3.47$, $SD = 1.65$) or previously married ($M = 3.44$, $SD = 1.63$) is similar. (Perhaps neurotic people tend not to get married, or marriage results in less neuroticism.)

5) Provide a written summary of the information you generated in 3 and 4 above. When presenting your summary please note you will need to provide enough information about the variables so that someone not familiar with the data can understand.

Researchers collected data from 1799 participants regarding several variables, including their personality scores (along the “Big Five” traits: Openness, Conscientiousness, Extroversion, Agreeableness, and Neuroticism), their educational background, community of residence (urban, suburban, or rural), age, race, and marital status (currently married, never married, or previously married).

The sample had an average age of 28.44 years ($Mdn = 23.0$, $SD = 13.76$). Regarding education, most of the sample were high school graduates ($n = 757$), followed by university graduates ($n = 488$), those who did not graduate high school ($n = 278$), those who attended and completed graduate school ($n = 245$), and those who did not provide information on this item ($n = 31$).

Most participants lived in suburban neighborhoods ($n = 815$), followed by urban neighborhoods ($n = 645$), or rural neighborhoods ($n = 317$). Twenty-two people did not specify their community of residence.

Most participants were white ($n = 1440$), followed by those who specified “other” ($n = 179$), those who were Asian ($n = 119$), Black ($n = 28$), or Arab ($n = 14$). Nineteen people did not specify their race.

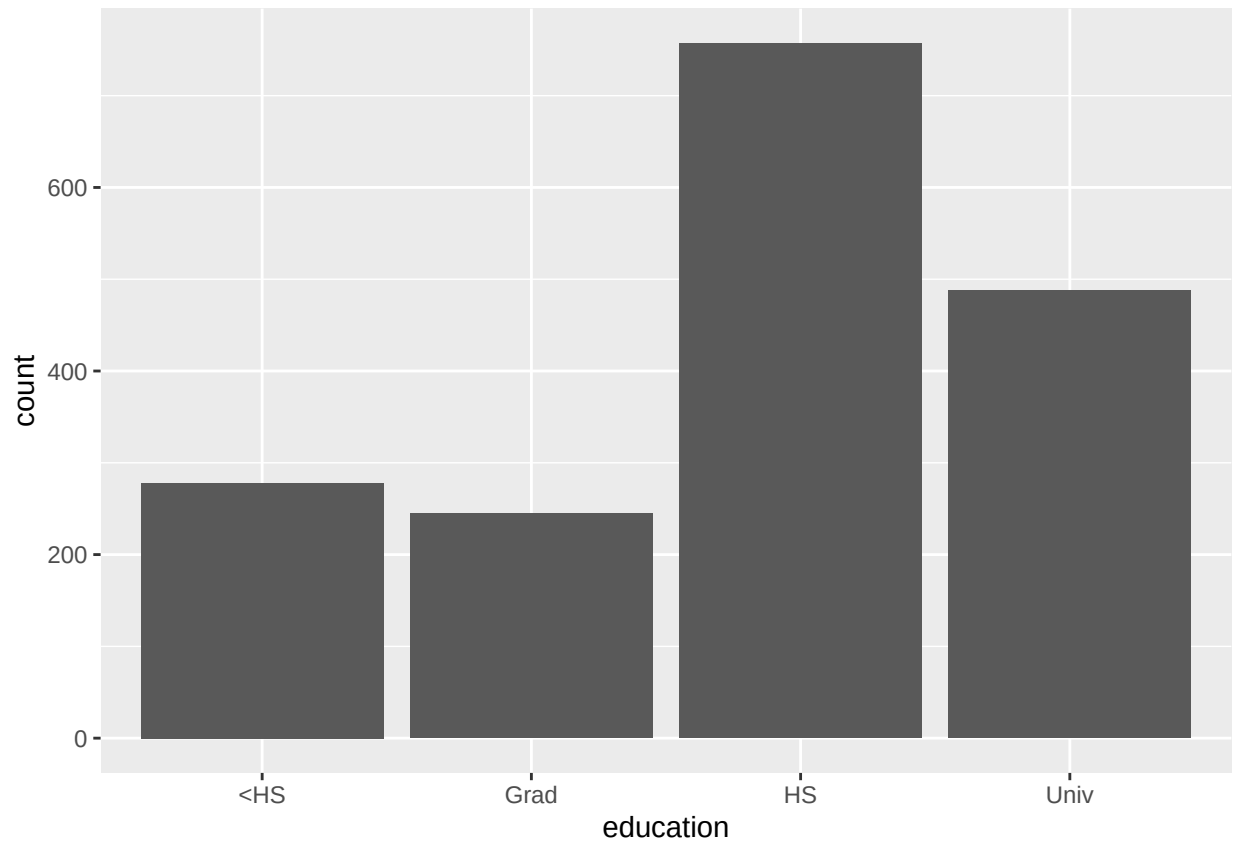
Participants were largely never married ($n = 1383$); 314 people were currently married, and 93 were previously married (divorced). Nine people did not specify their marital status.

Regarding personality characteristics, the sample had a mean extraversion of 3.20 ($SD = 1.68$), mean neuroticism of 3.97 ($SD = 1.76$), mean conscientiousness of 4.39 ($SD = 1.56$), mean agreeableness of 4.30 ($SD = 1.41$), and mean openness of 5.50 ($SD = 1.21$).

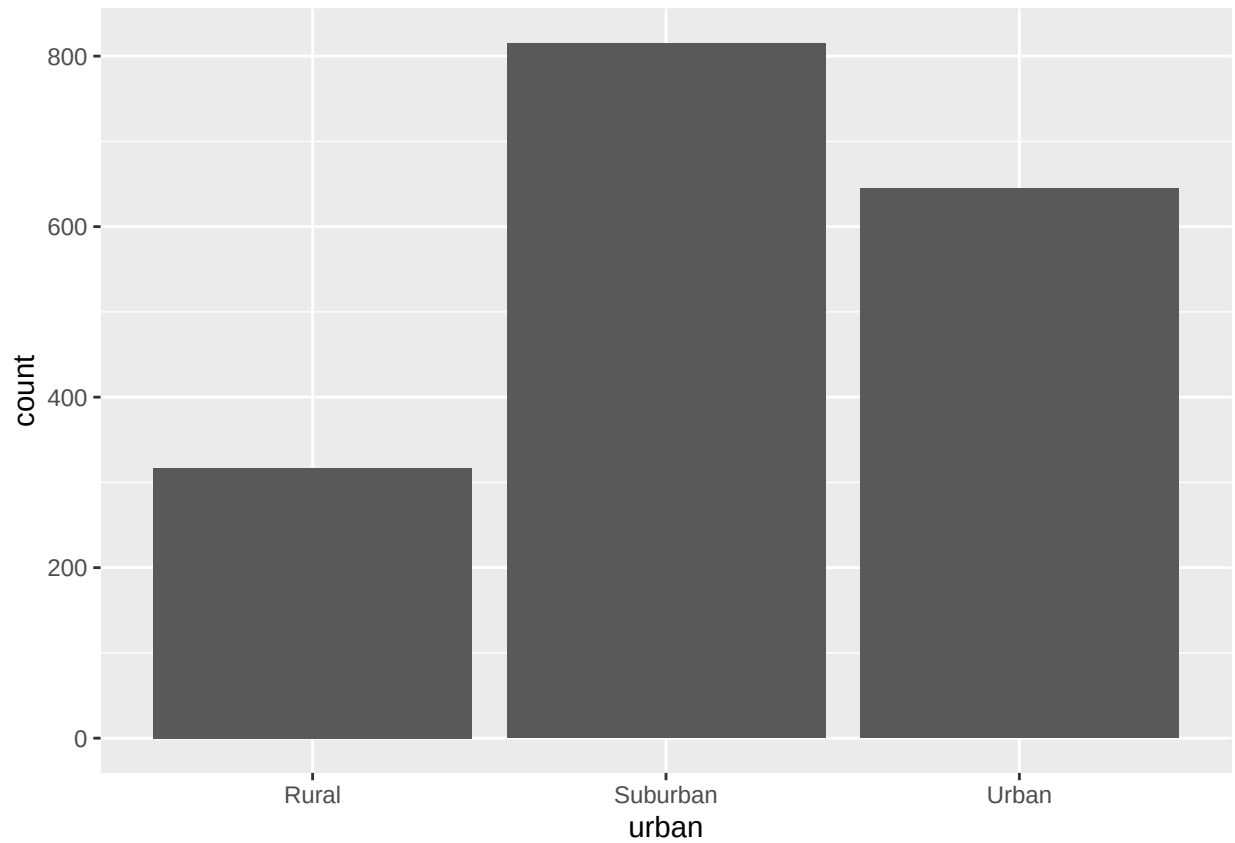
The plots below summarise the descriptive statistical information above.

(NOTE: In APA, you should actually label and format each plot differently so it is neater, and each time you refer to a plot, it’s with a purpose. Doing this kind of thing in actual papers is never a good idea, unless you put it in the supplemental. We usually avoid plotting descriptive stats beyond EDA, and opt instead for an informative table (or tables) for numeric descriptives.)

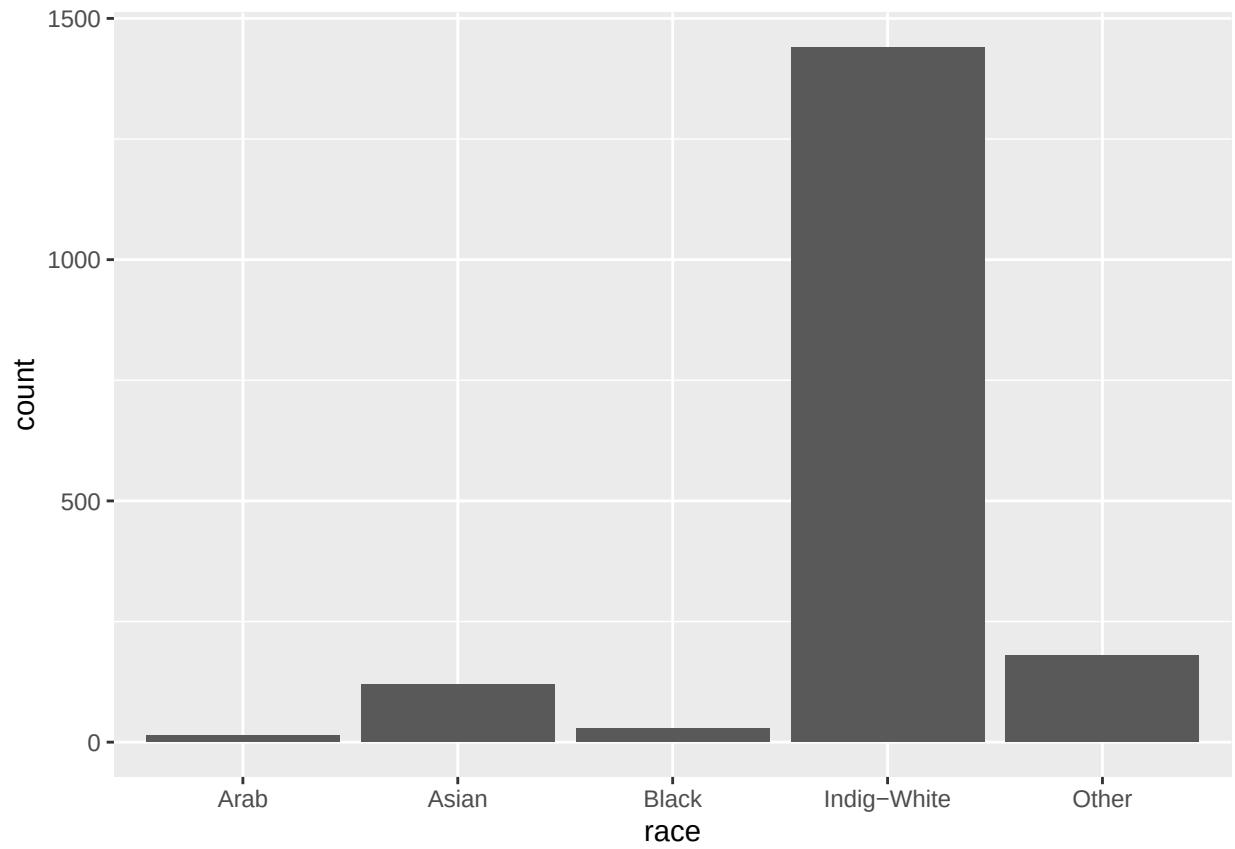
```
personality_final |>
  filter(!is.na(education)) |>
  ggplot(aes(x = education)) +
  geom_bar()
```



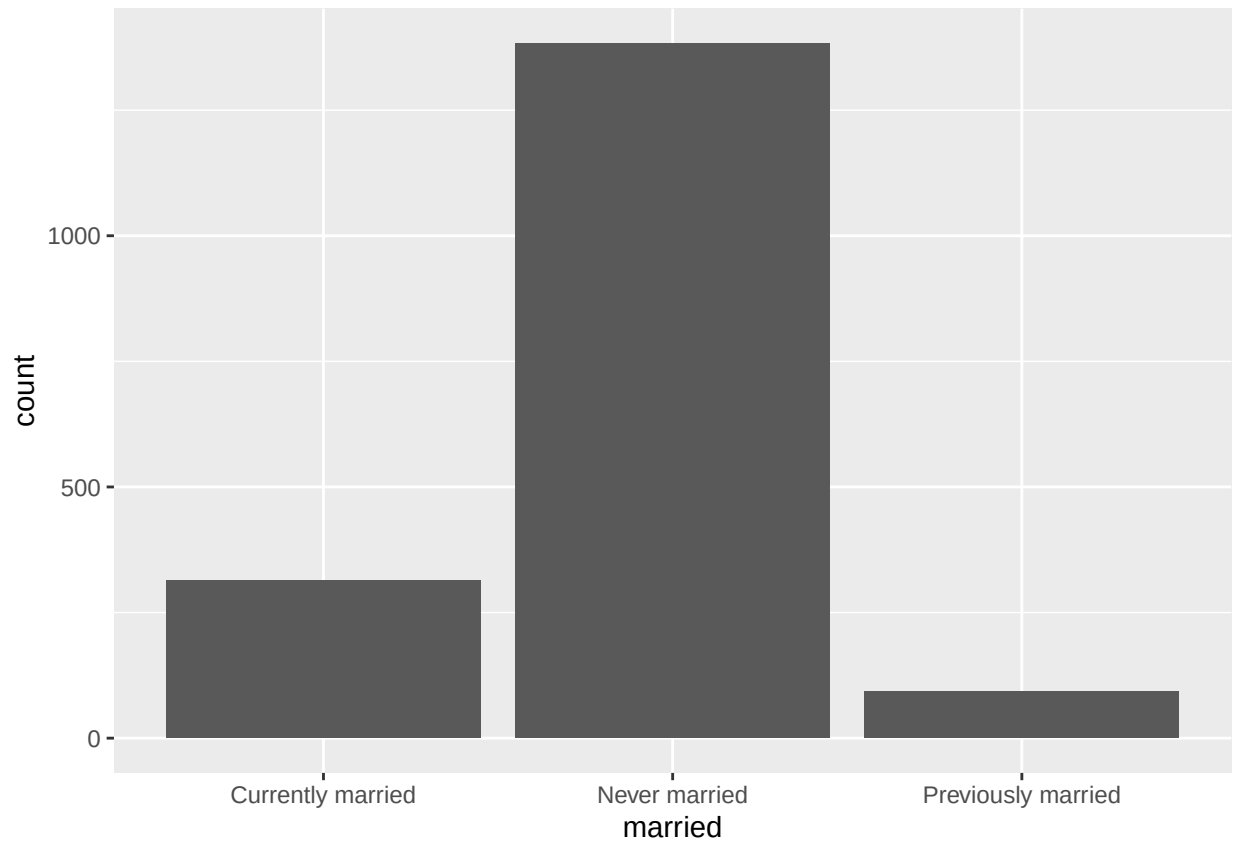
```
personality_final |>  
  filter(!is.na(urban)) |>  
  ggplot(aes(x = urban)) +  
  geom_bar()
```



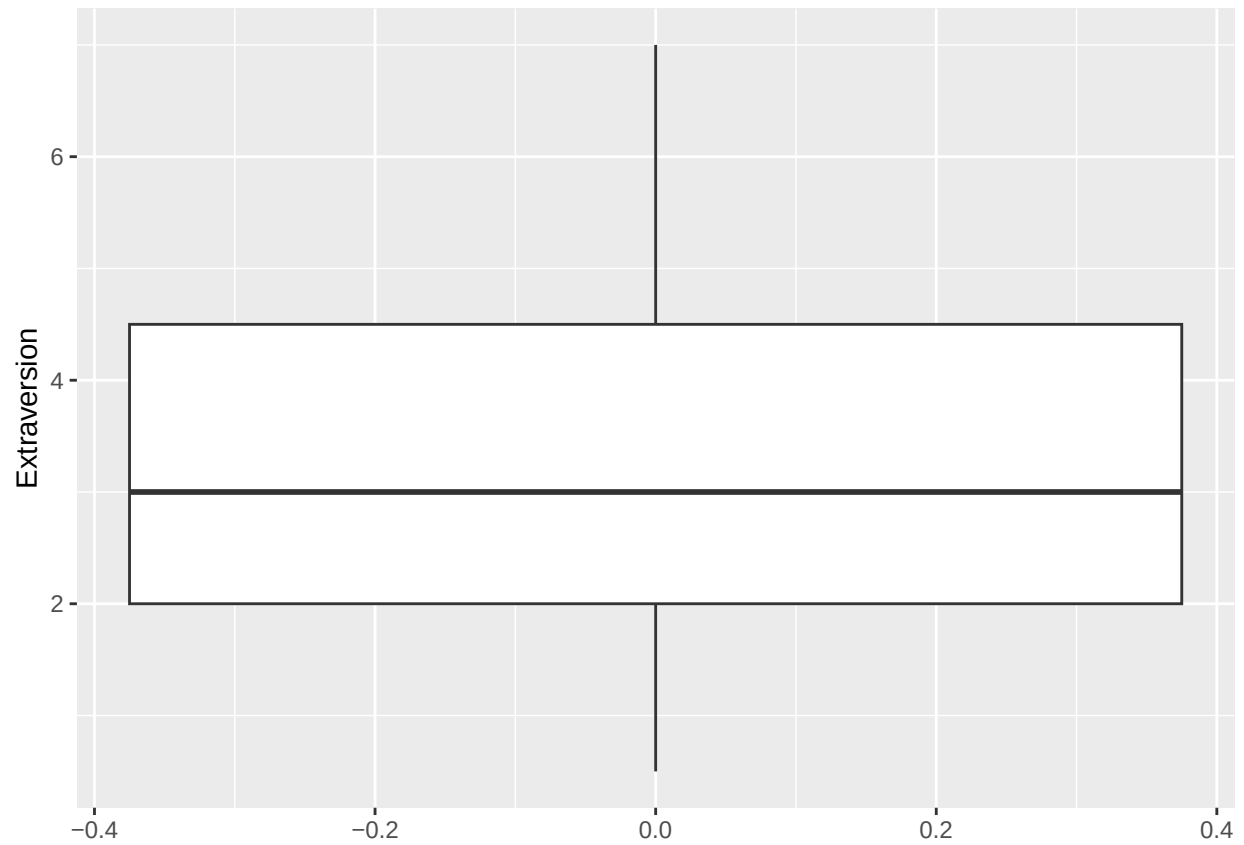
```
personality_final |>  
  filter(!is.na(race)) |>  
  ggplot(aes(x = race)) +  
  geom_bar()
```



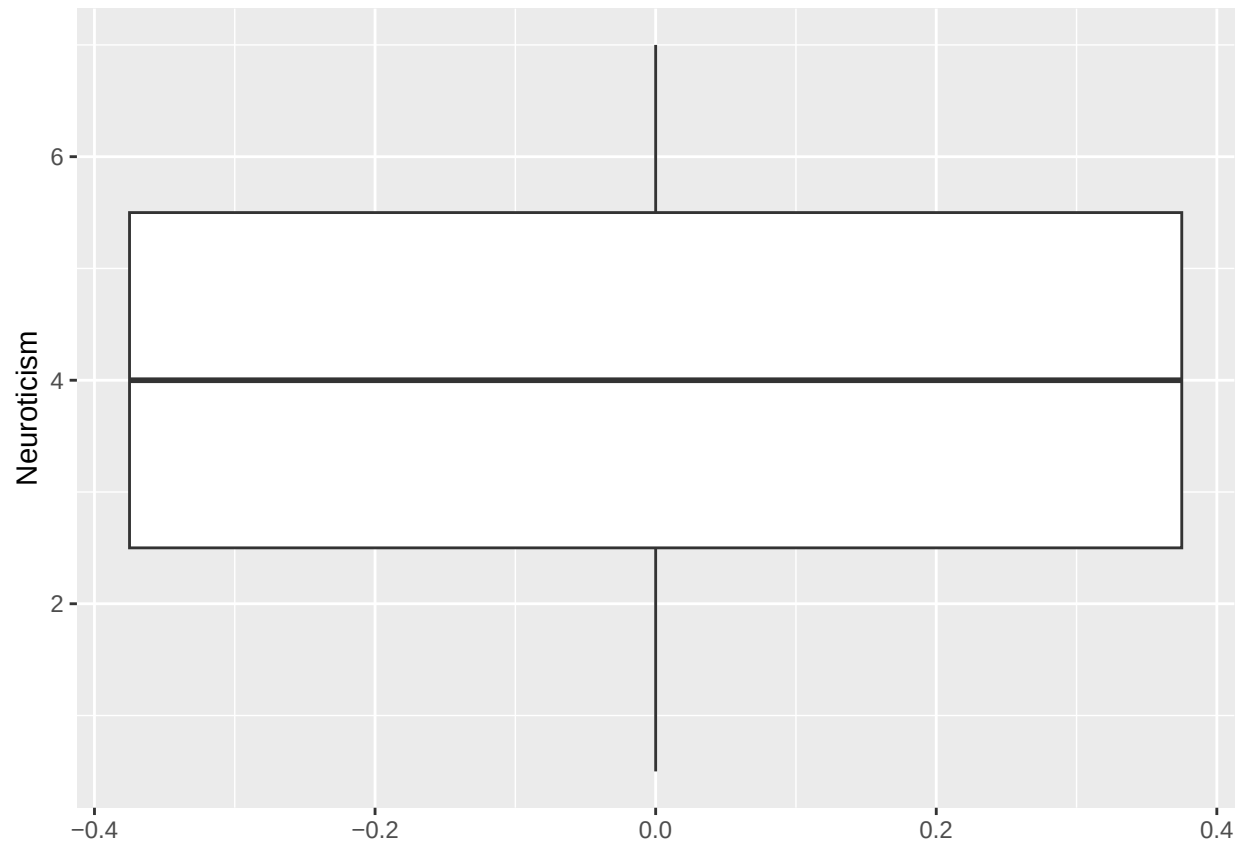
```
personality_final |>  
  filter(!is.na(married)) |>  
  ggplot(aes(x = married)) +  
  geom_bar()
```



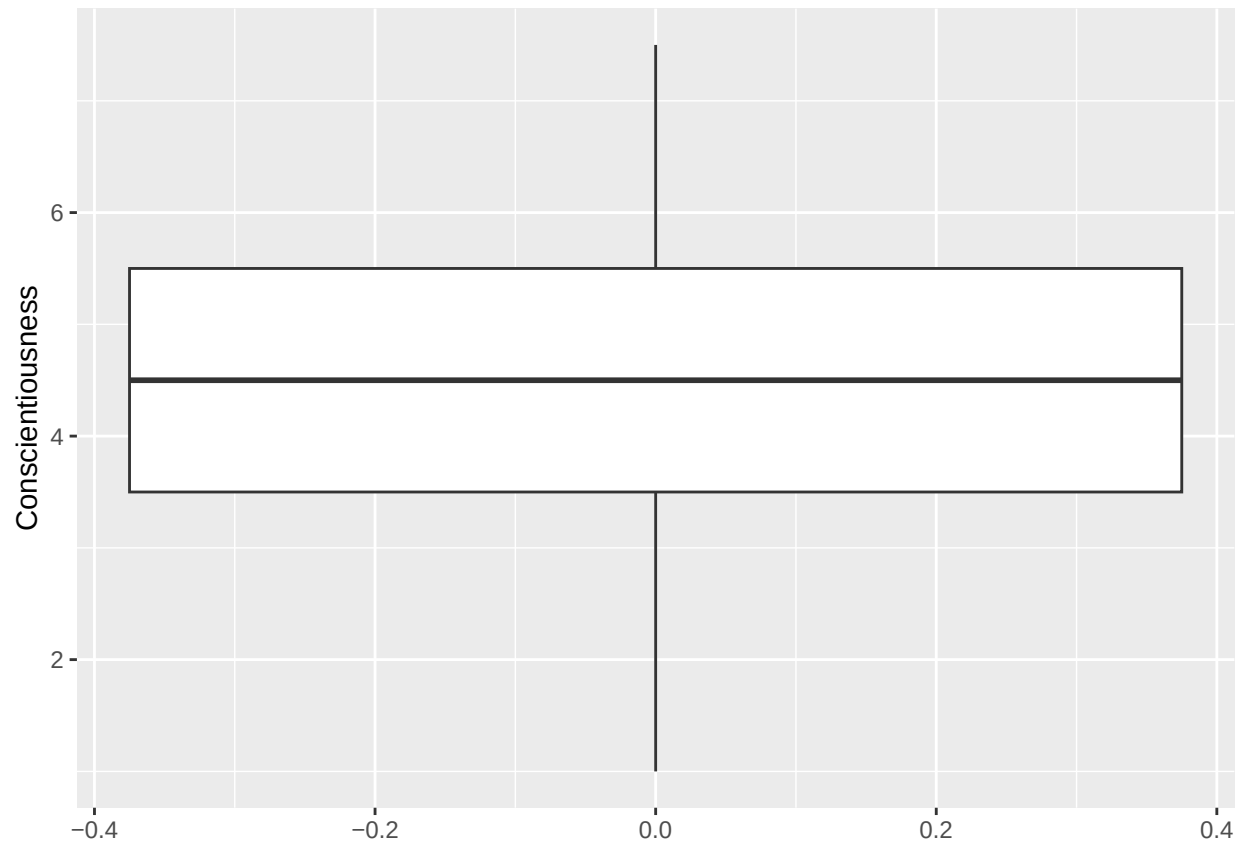
```
personality_final |>  
  ggplot(aes(y = Extraversion)) +  
  geom_boxplot()
```



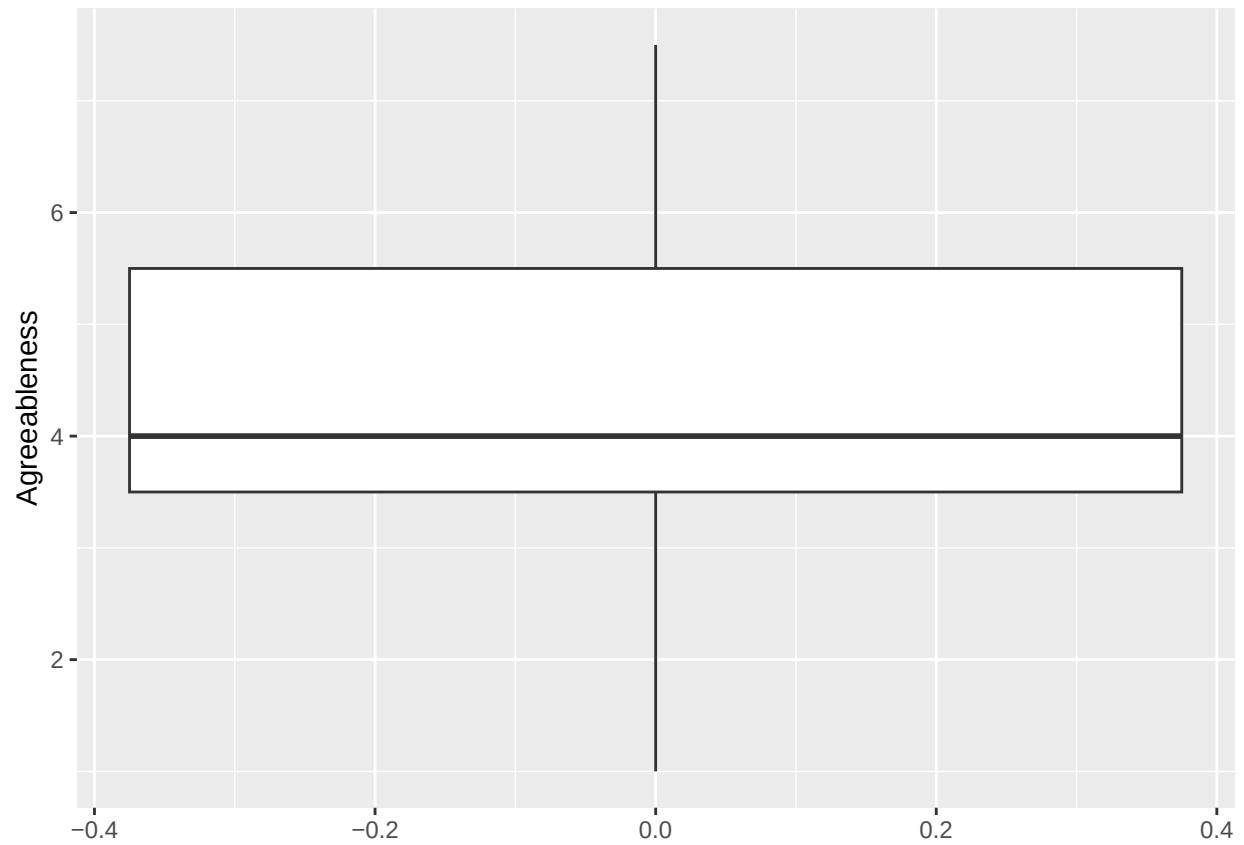
```
personality_final |>  
  ggplot(aes(y = Neuroticism)) +  
  geom_boxplot()
```



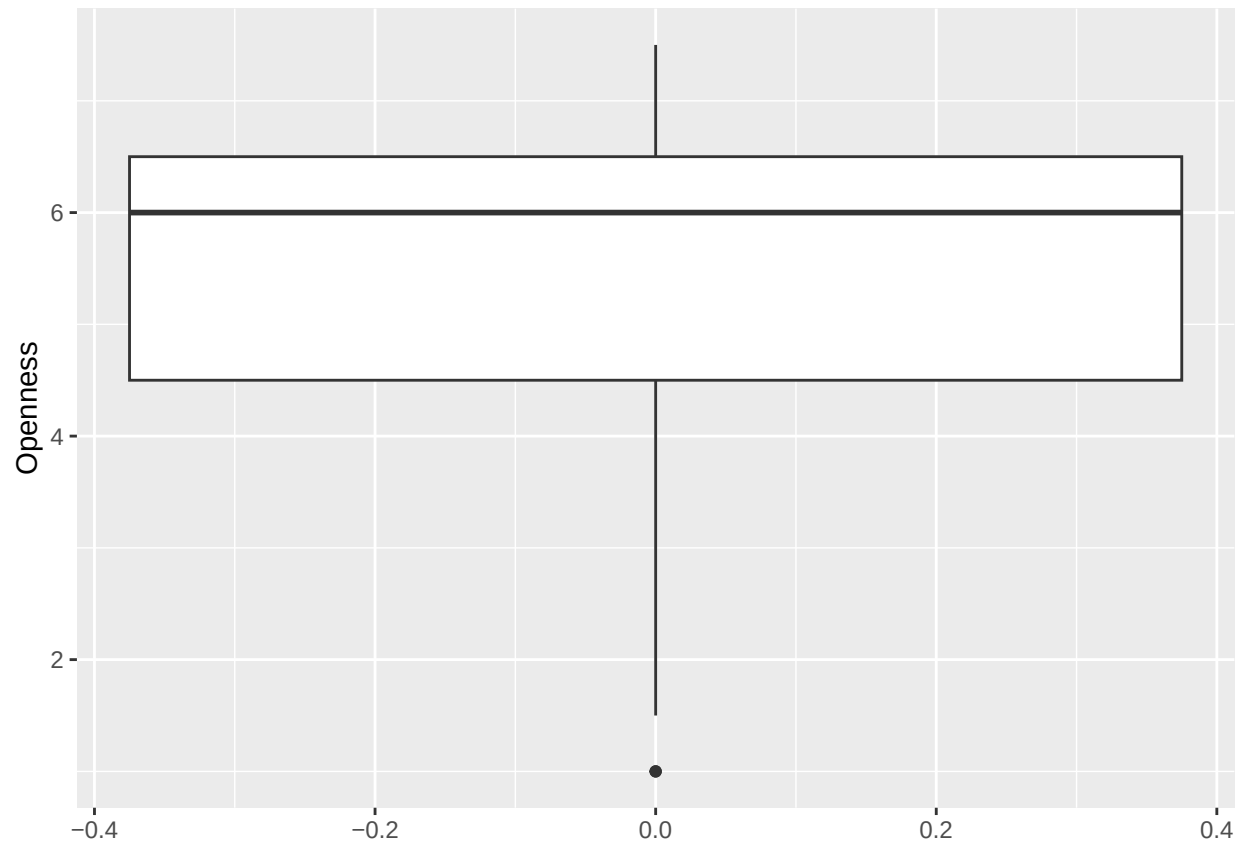
```
personality_final |>  
  ggplot(aes(y = Conscientiousness)) +  
  geom_boxplot()
```



```
personality_final |>  
  ggplot(aes(y = Agreeableness)) +  
  geom_boxplot()
```

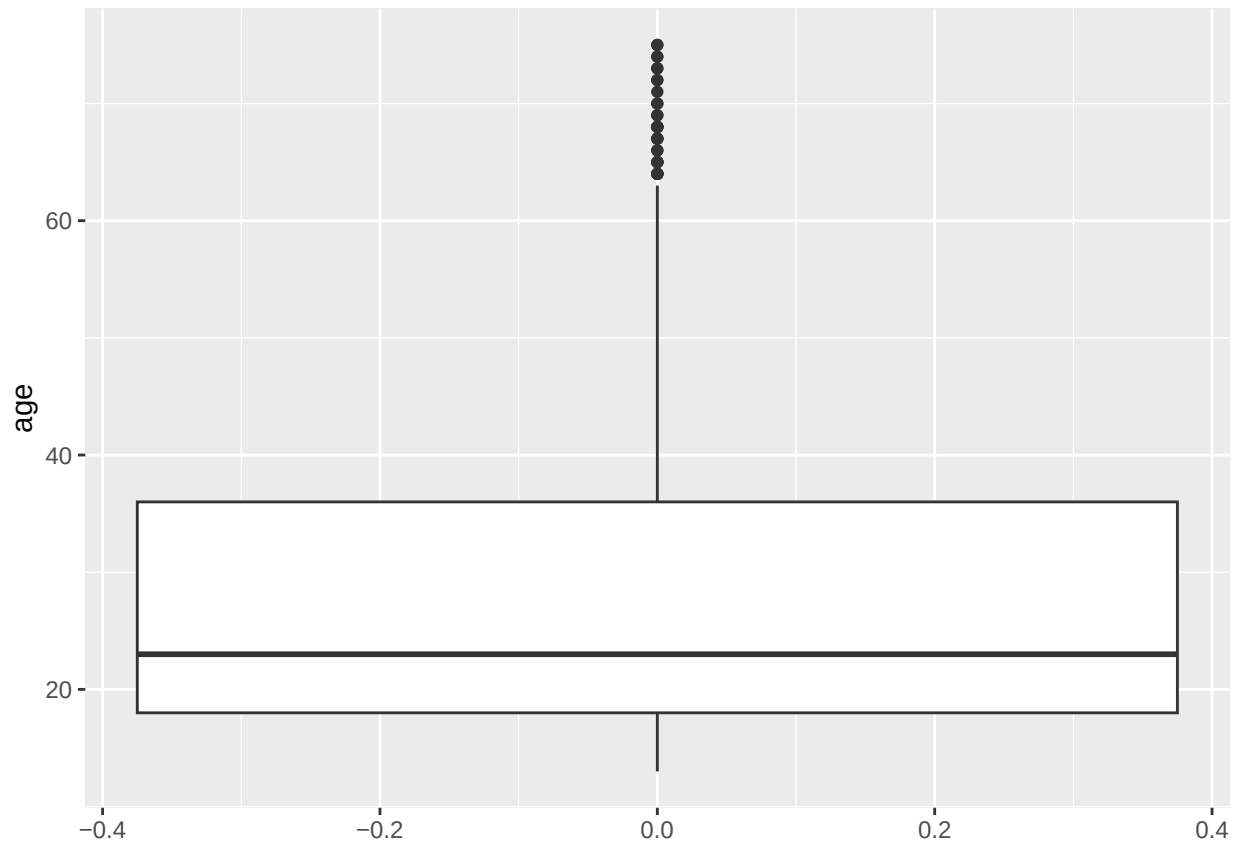


```
personality_final |>  
  ggplot(aes(y = Openness)) +  
  geom_boxplot()
```



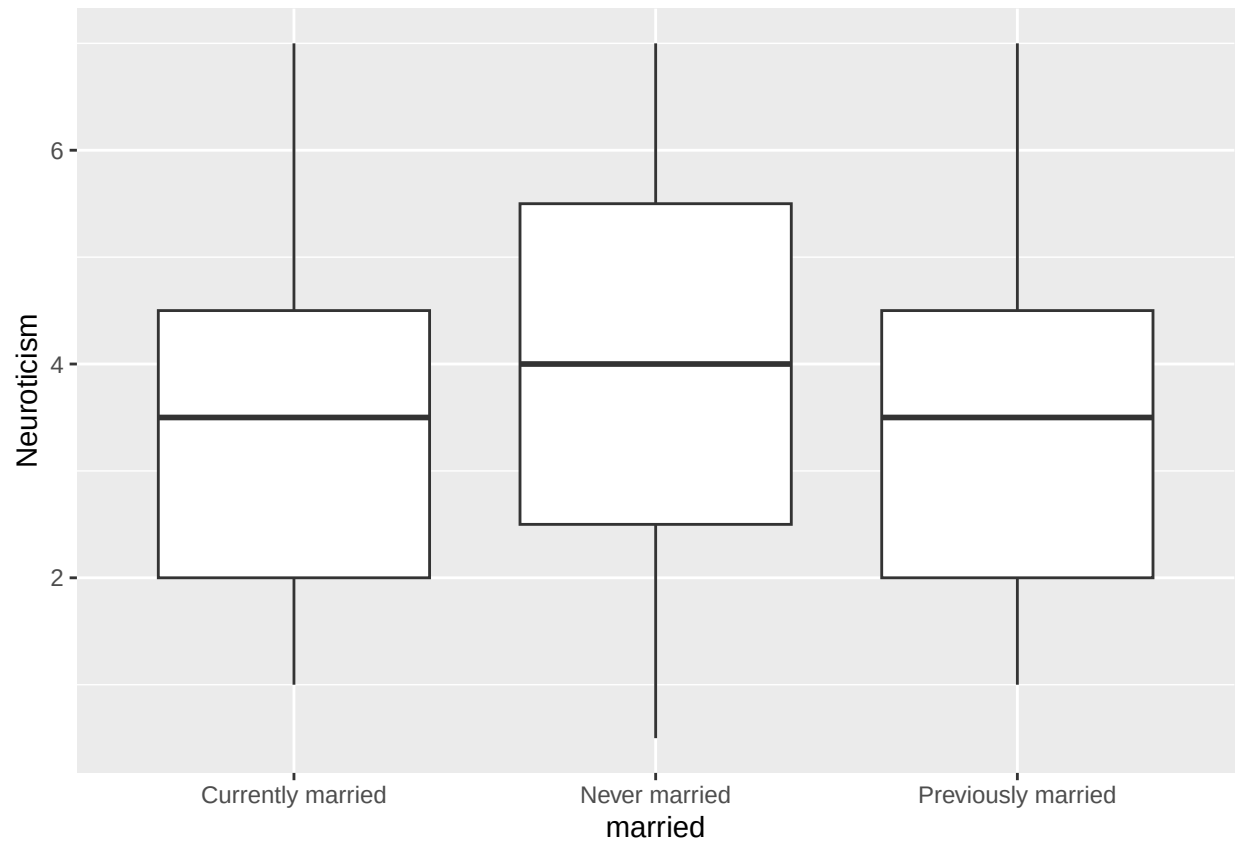
```
personality_final |>  
  ggplot(aes(y = age)) +  
  geom_boxplot()
```

```
## Warning: Removed 1 row containing non-finite outside the scale range  
## (`stat_boxplot()`).
```



I was interested in determining if there was a mean difference in neuroticism between those of different marital statuses. The plot below summarises the mean difference between those who have never been married, those who were once married, and those who are currently married.

```
personality_final |>  
  # Filter NAs for clearer plot  
  filter(!is.na(married)) |>  
  ggplot(aes(x = married, y = Neuroticism)) +  
  geom_boxplot()
```



As the above plot shows, the mean neuroticism ($M = 4.12$, $SD = 1.77$) is highest within those who are never married. The mean for those who are currently married ($M = 3.47$, $SD = 1.65$) or previously married ($M = 3.44$, $SD = 1.63$) is similar.